Quantification of the uncertainty in machine learning model predictions for biologocal profile estimation problems

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Subject
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ETSIIT
Escuela Técnica Superior
de Ingenierías Informática
y de Telecomunicación

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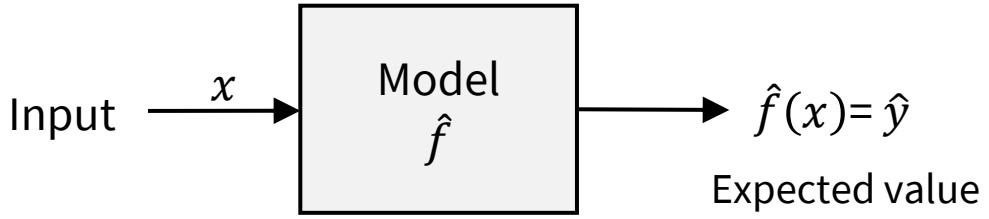
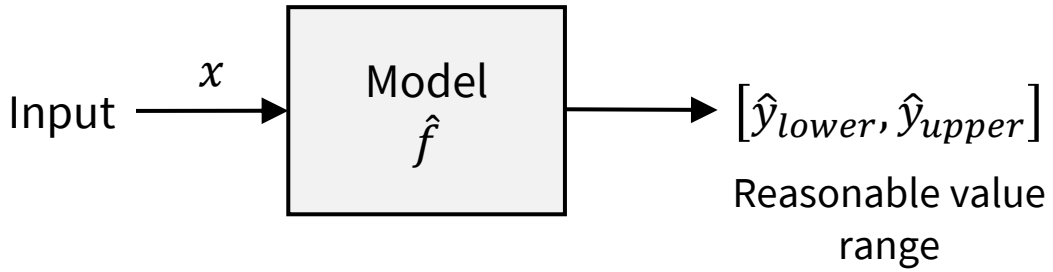
Problem description

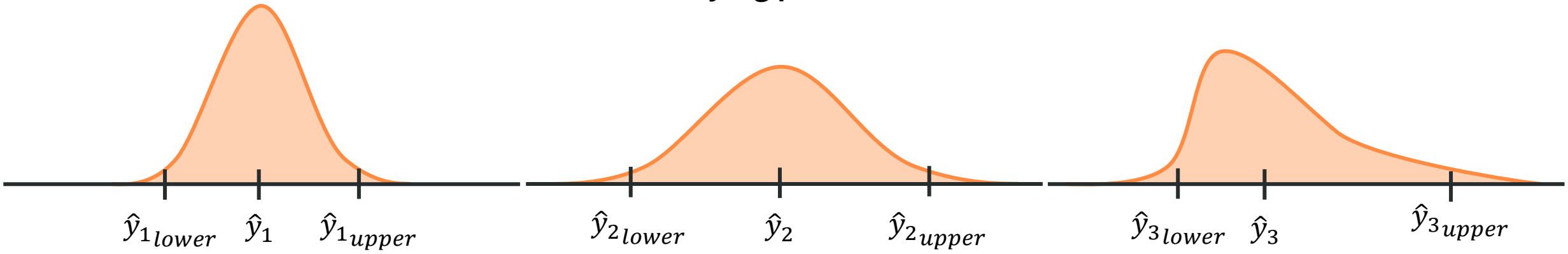
Dominant paradigm: Point prediction



Alternativa: Predicción interválica



Underlying phenomenon



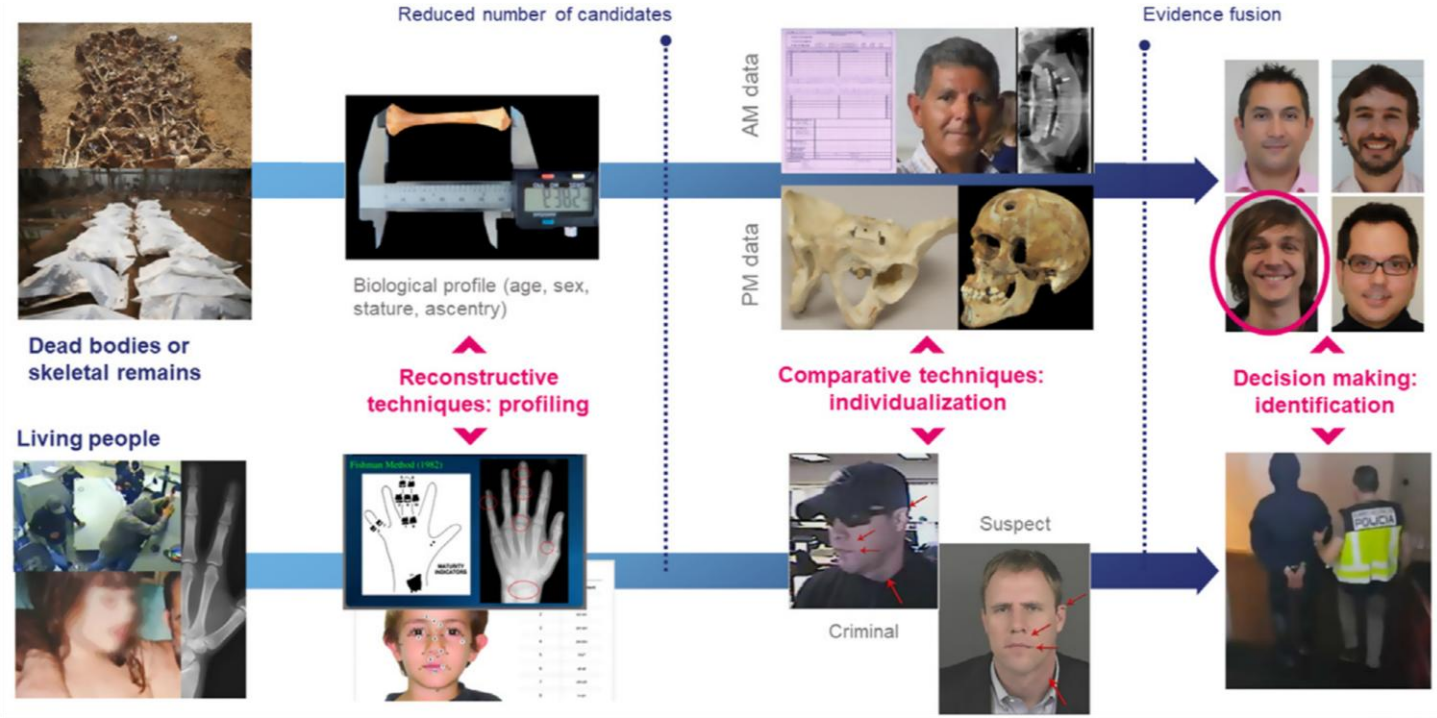
Problem background (I)

▶ Forensic Anthropology

Techniques in physical anthropology for resolving **medico-legal issues**.

Biological Profile

- **Sex**
- **Age**
- **Height**
- **Ancestry**



Forensic identification based on the human skeleton [1].

[1] Mesejo, P., Martos, R., Ibáñez, Ó., Novo, J., & Ortega, M. (2020). A survey on artificial intelligence techniques for biomedical image analysis in skeleton-based forensic human identification. *Applied Sciences*, *10*(14), 4703.

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▶ Biological Profile

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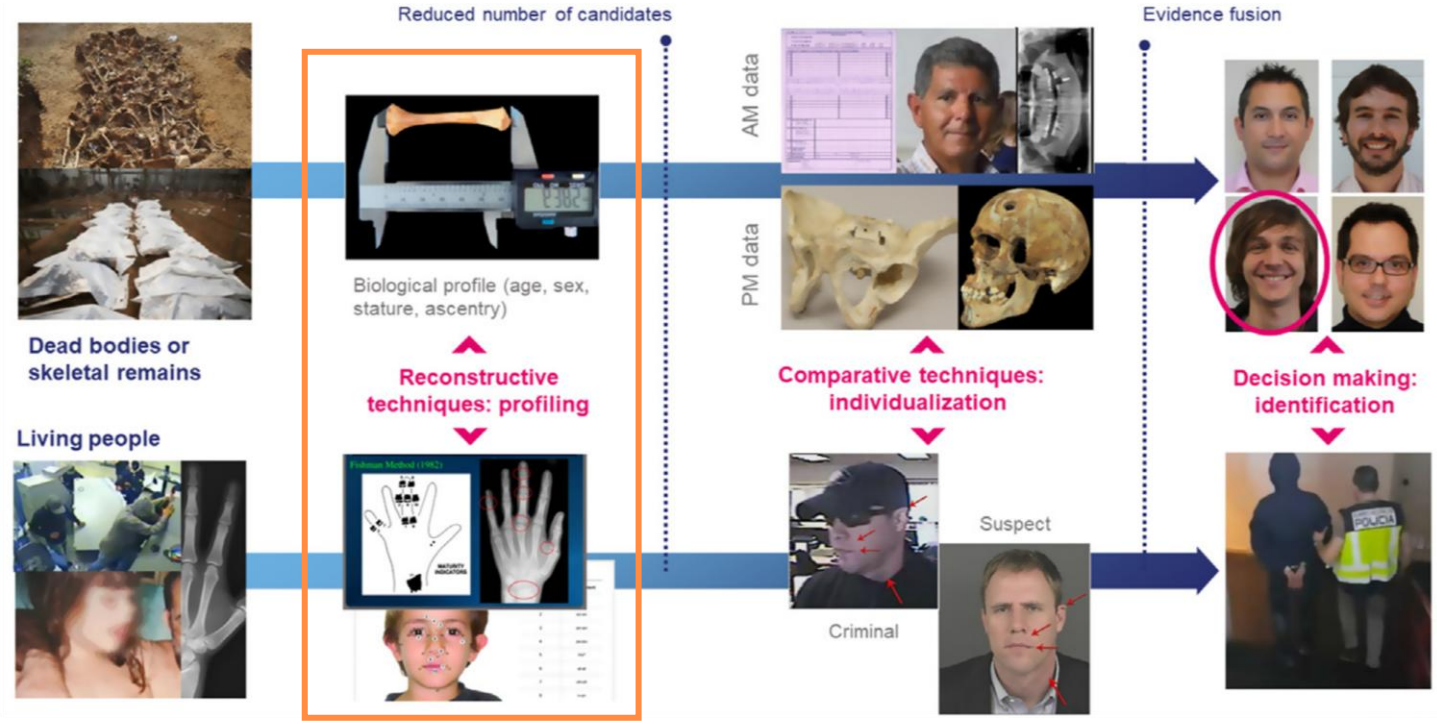

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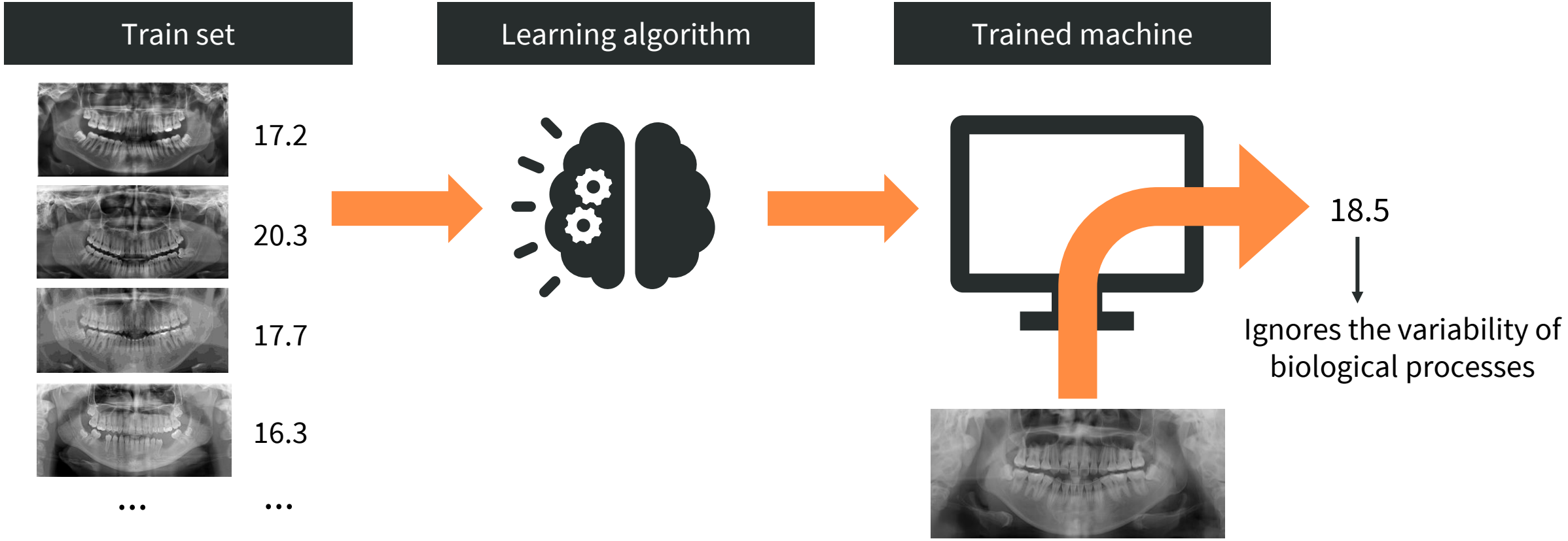
Contexto del problema (II)

- **Uso of AI techniques and *machine learning*** → Standardizes and enhances objectivity in the assessment of biological profiles



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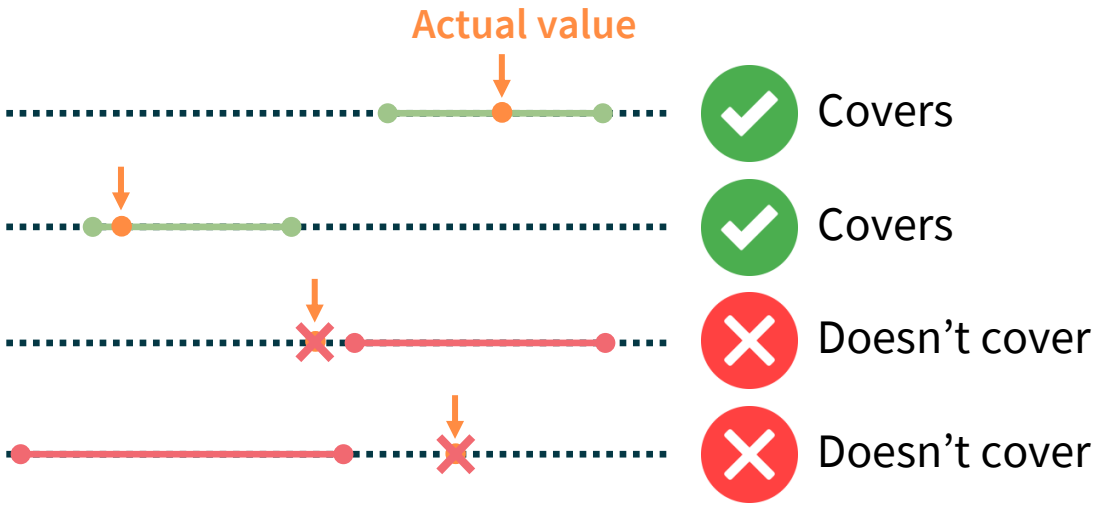
Conformal prediction (I)

Theoretical framework for generating prediction intervals with statistical coverage guarantees for a desired confidence level.



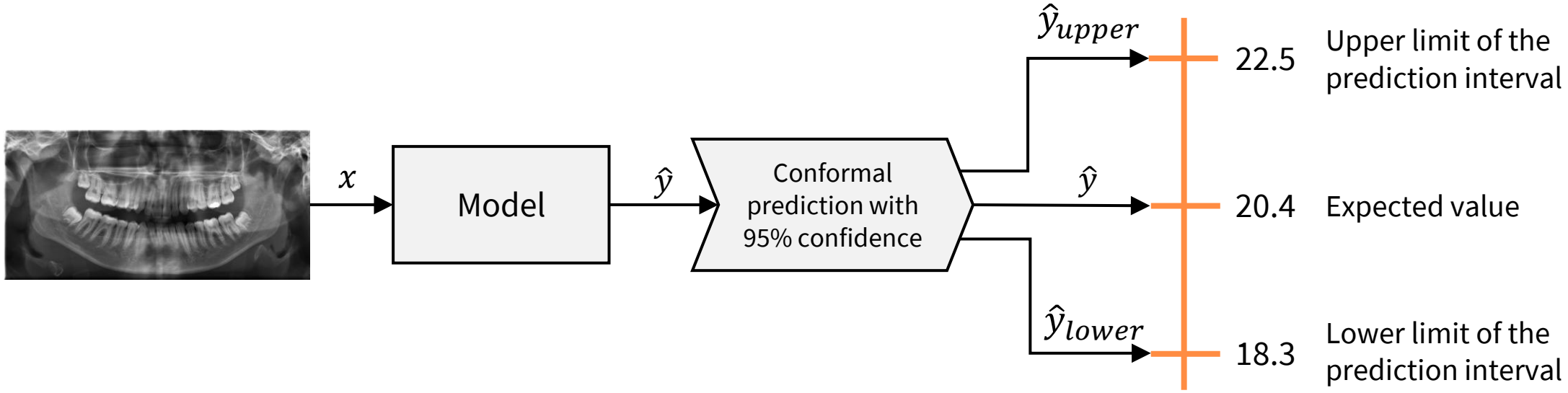
90 % confidence

statistically guarantees

90% of the intervals cover the true value

- **Independent** of the data **distribution**. It only **assumes** data **interchangeability**.
- Model-independent (***model-agnostic***).
- **Calibration:** New step in model preparation, immediately following training.
- Requires a **representative calibration dataset** for effective coverage.

Conformal prediction (II)



Objectives

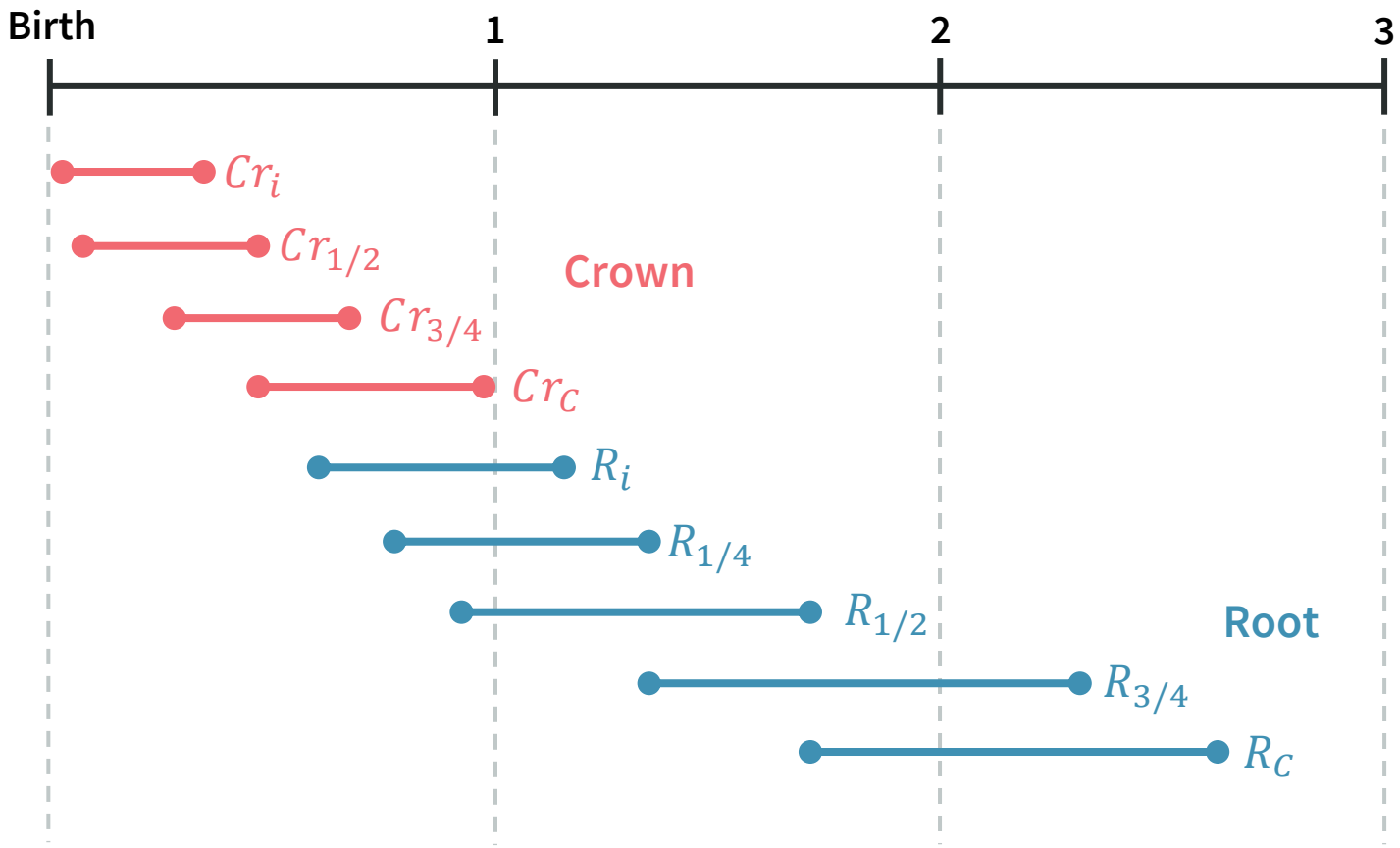
1. **Demonstrate the importance of uncertainty quantification** compared to error analysis in point predictions

2. **Apply point and interval methods** to a practical context such as age estimation

3. **Compare and discuss results**

Quantifying uncertainty in age estimation using *machine learning*

Age ranges used for assessing indicators cover 95% of cases



Excerpt from the dental training schedule for women. Retrieved from [1].

[1] Byers, S. N., & Juarez, C. A. (2023). *Introduction to forensic anthropology*. Routledge.

Quantifying uncertainty in age estimation using *machine learning*

Most commonly used metrics:

- **Mean ± Standard deviation**
- **Error histogram**

By gender

	cnn-2d-bones-carpal	cnn-2d-hand
	(mean ± std [years])	**(mean ± std [years])**
	error histogram [years] (|-3|-2|-1|0|+1|+2|+3|)	**error histogram [years] (|-3|-2|-1|0|+1|+2|+3|)**
Male	**0.49 ± 0.60** |1|13|66|238|92|8|0|	**0.58 ± 0.64** |2|14|97|207|86|10|2|
Female	**0.66 ± 0.61** |0|7|80|174|132|24|0|	**0.89 ± 0.75** |2|41|108|135|95|29|7|
All	**0.57 ± 0.61** |1|20|146|412|224|32|0|	**0.73 ± 0.72** |4|55|205|342|181|39|9|

Excerpt from the error study for the proposed 2D methods in [1]

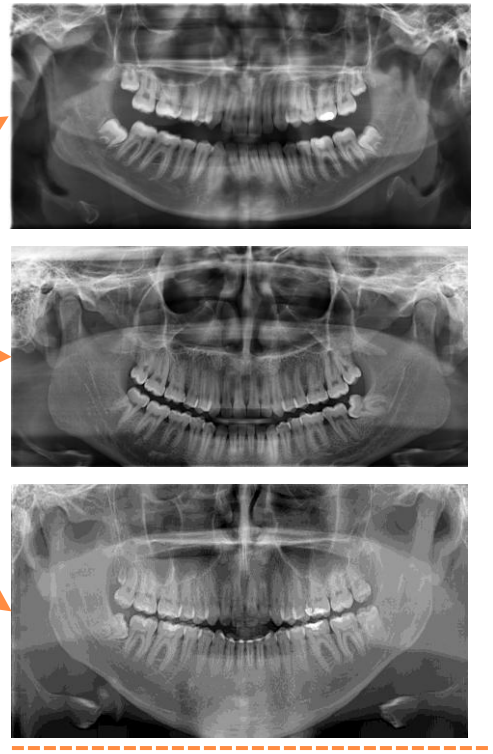
[1] Štern, D., Payer, C., & Urschler, M. (2019). Automated age estimation from MRI volumes of the hand. Medical image analysis, 58, 101538.

Data set (I)

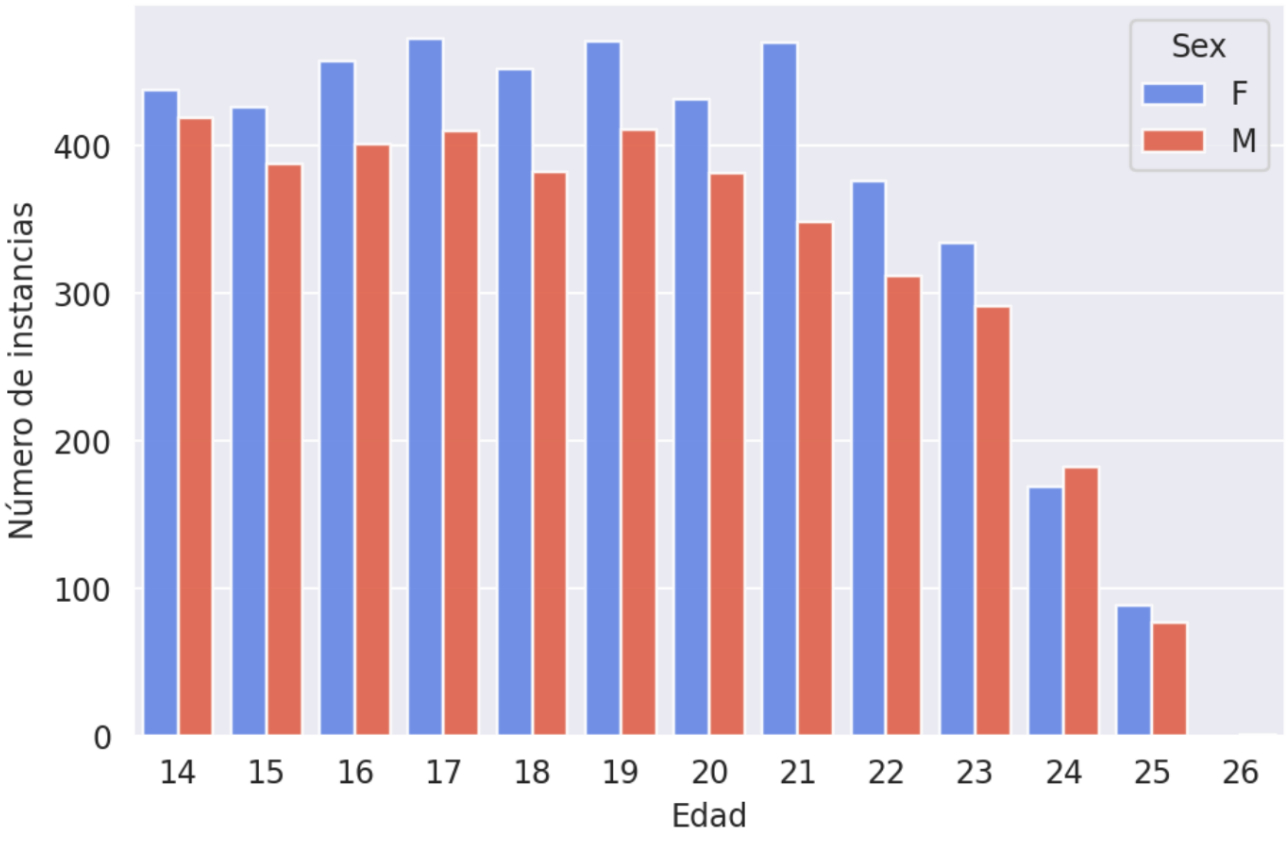
- 10 739 samples

Orthopantomograph (OPT)

Chronological Age	Sex	OPT
25.32	M	ID1
19.46	F	ID2
23.71	M	ID3
...	...	...



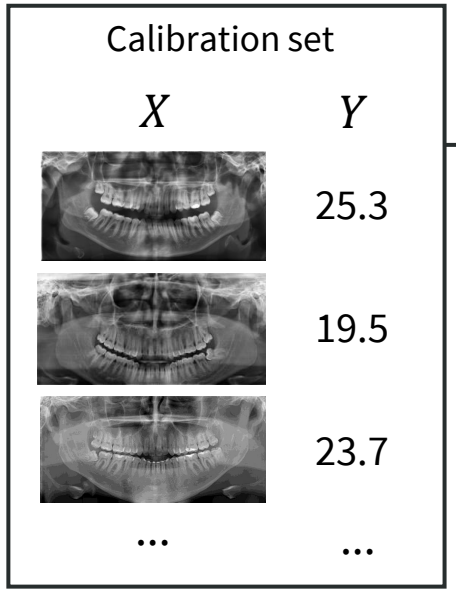
Data set (II)



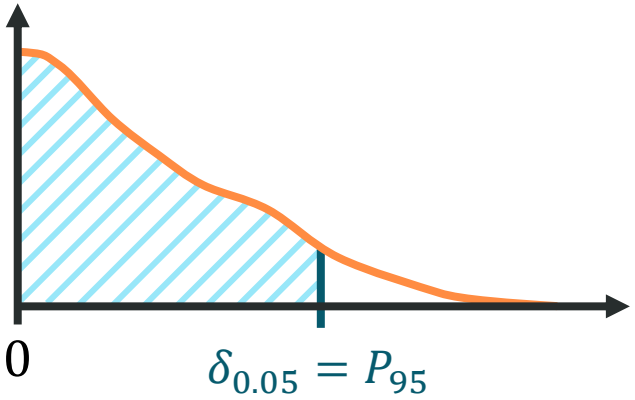
- Ages between 14 and 25
- Data imbalanced by age: decrease in samples starting from age 22/23.
- Moderate imbalance in the data by gender

Methods for interval prediction (I)

Inductive Conformal Prediction (ICP)



$\{x_i\}_1^n$ → Model $\hat{f}$ → $\{\hat{y}_i\}_1^n$ → Absolute error $|y_i - \hat{y}_i|$



Non-conformity threshold

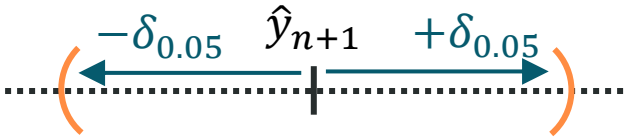
- Provides **coverage guarantees**
- **Model-agnostic**
- **Simetric Interval** and **fixed width**

New

x_{n+1} → Model $\hat{f}$ → $\hat{y}_{n+1}$ → ICP with 95% confidence

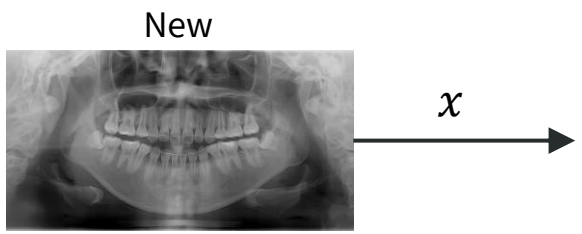
$[\hat{y}_{n+1} - \delta_{0.05}, \hat{y}_{n+1} + \delta_{0.05}]$

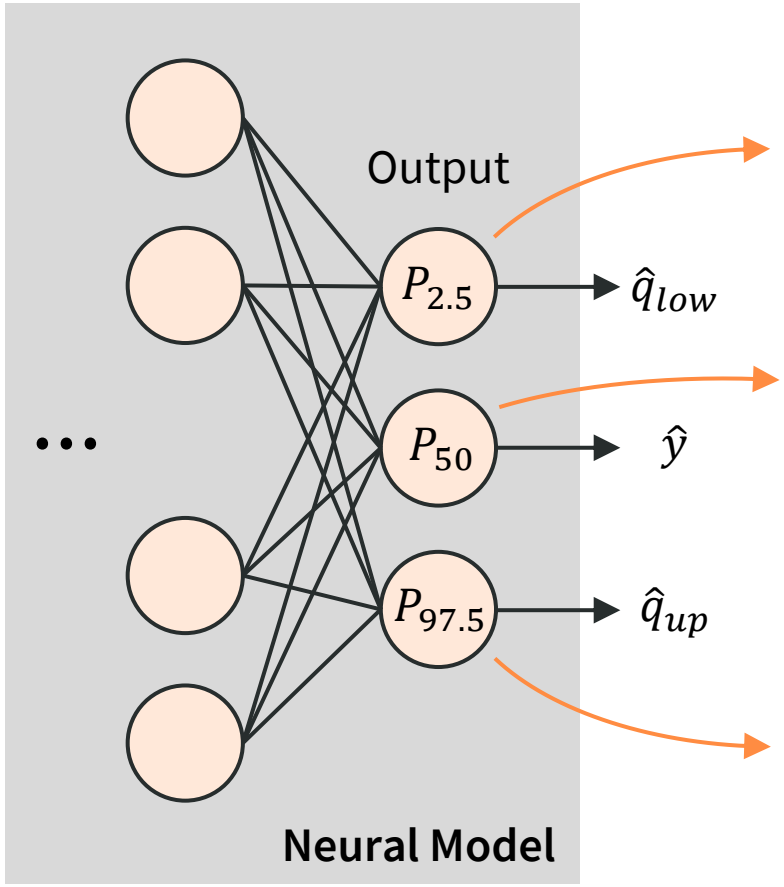
$\hat{y}_{n+1}$



Methods for interval prediction (II)

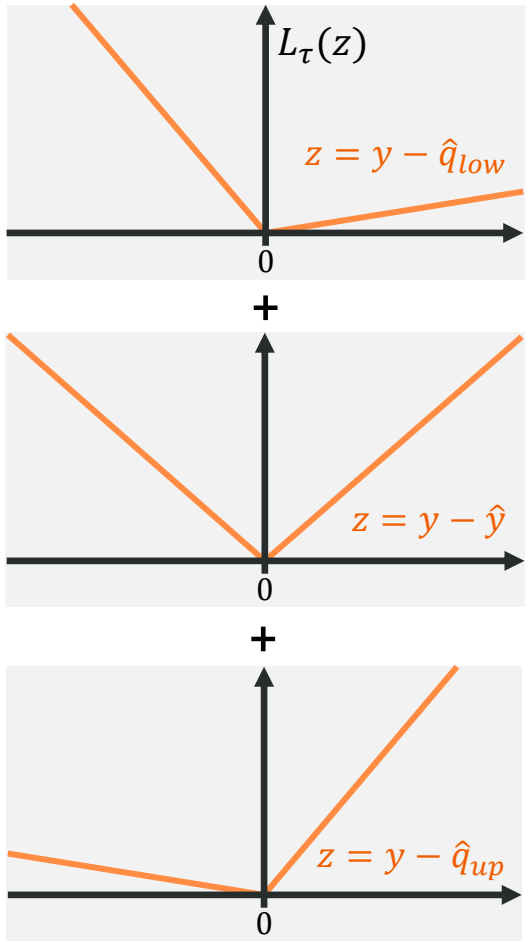
Quantile Regression (QR)





- **Coverage** is **not garanteed**
- **Model-dependent**
- **Simetric** interval and **variable width**

Loss function



Penalizes negative errors more heavily

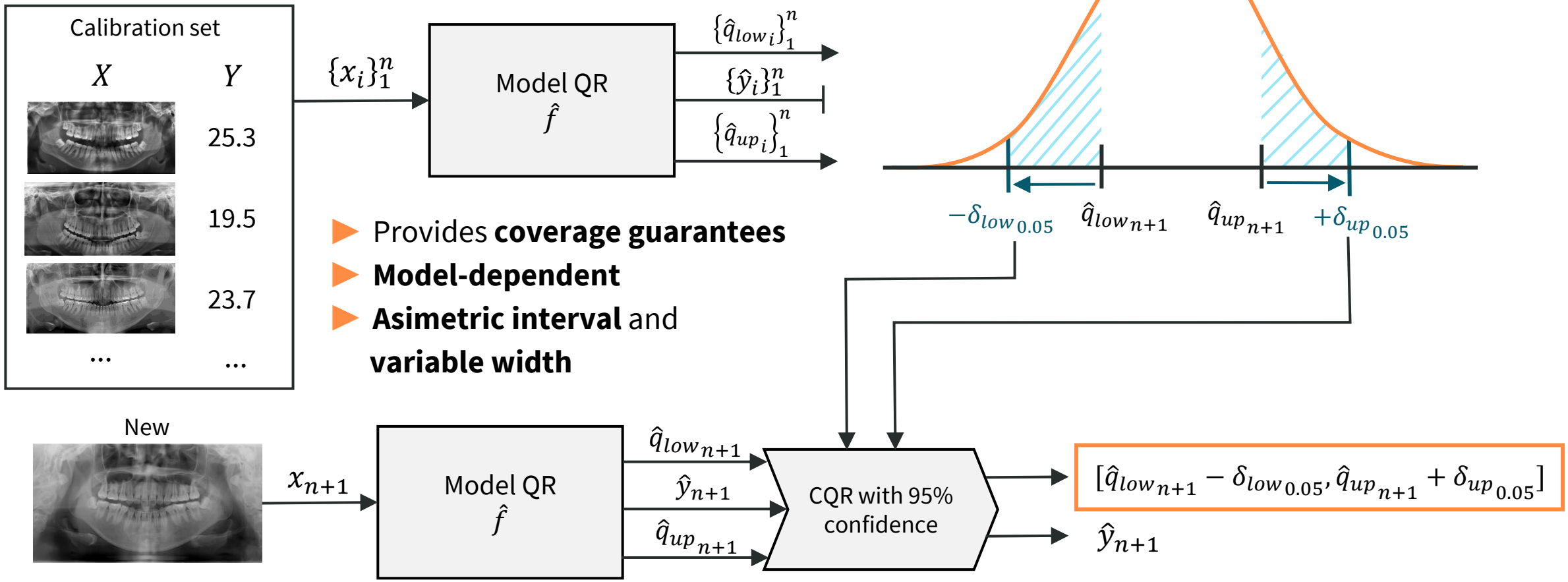
Penalizes both negative and positive errors equally

Penalizes positive errors more heavily

*

Methods for interval prediction (III)

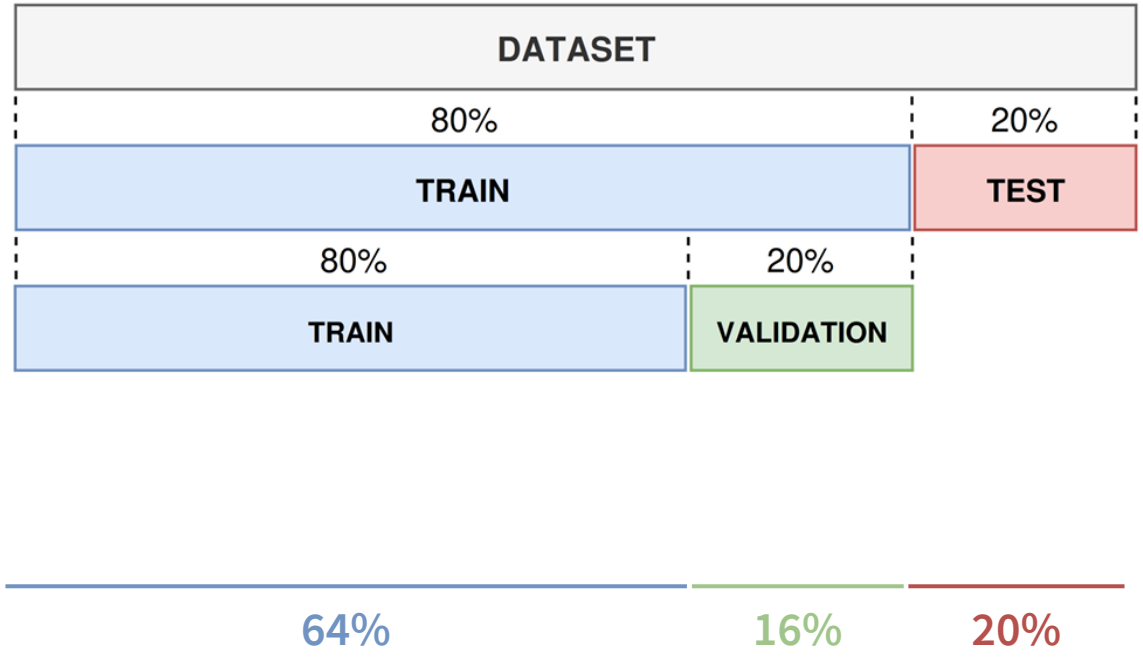
Conformalized Quantile Regression (CQR)



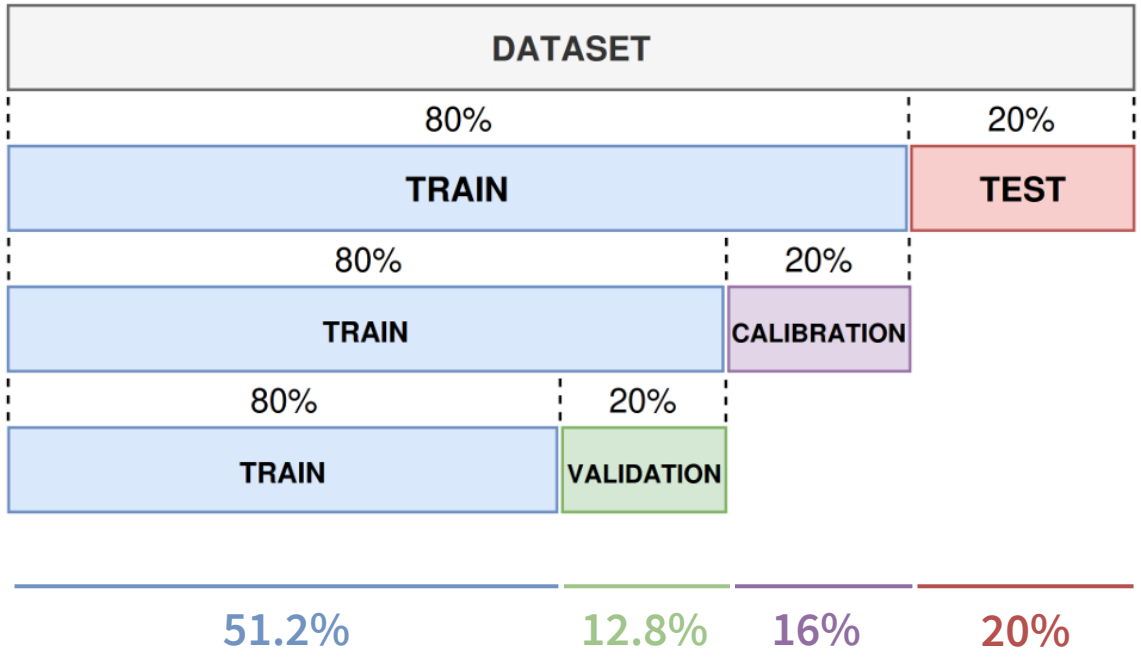
- Provides **coverage guarantees**
- **Model-dependent**
- **Asimetric interval** and **variable width**

Experimental validation protocol

Non-conformal methods:



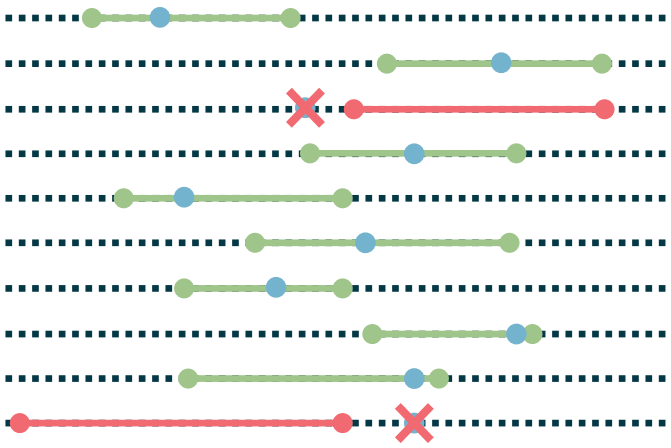
Conformal methods:



Metrics

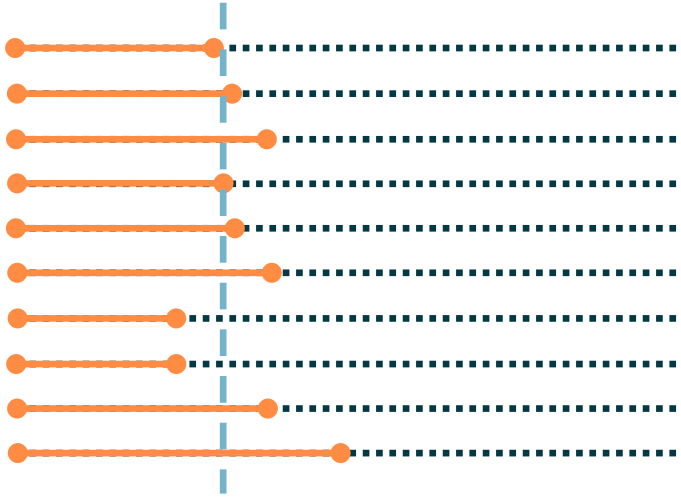
1 Empirical Coverage

Proportion of generated intervals that contain the actual value.



2 Average Interval Width

Average width of the generated prediction intervals.



They can be measured over the entire population or within subpopulations

Experimental design

1. Method base	2. Method ICP	3. Method QR	4. Method CQR
Point prediction ± 2 Average absolute error	95% confidence	95% confidence	95% confidence

Backbone: ResNeXt50

10 runs per method

Metrics obtained from testing

Comparative summary of the methods

Methods	base	ICP	QR	CQR
Coverange Guarantee	✗	✓	✗	✓
Requires calibration	✗	✓	✗	✓
Adaptive intervals (variable size)	✗	✗	✓	✓
Model-agnostic	✓	✓	✗	✗

Results (I)

Point regression metrics

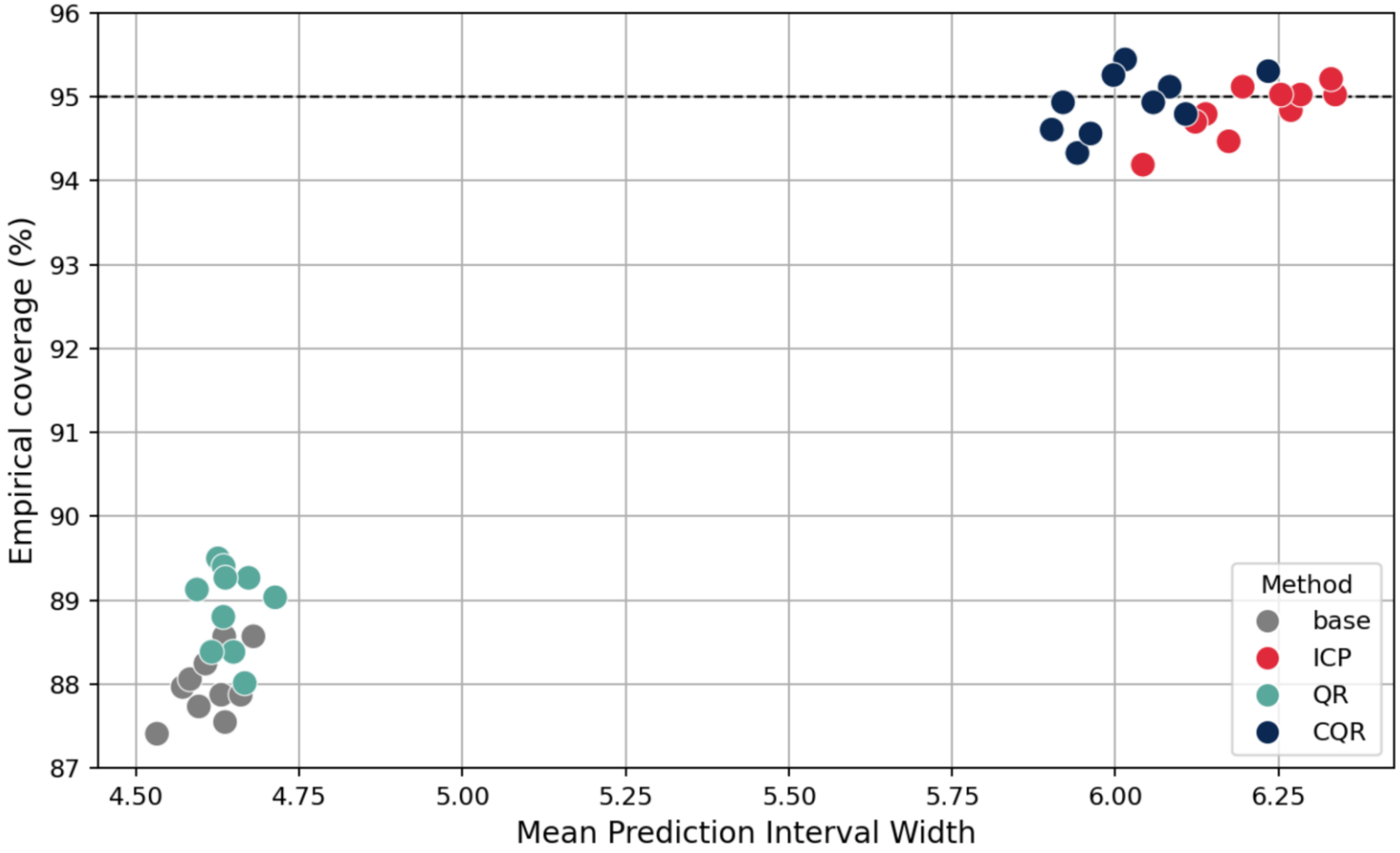
- Conformal methods result in minimal deterioration of the metrics.

Run	Mean Absolute Error			
	base	ICP	QR	CQR
Run 1	1.17	1.20	1.17	1.18
Run 2	1.15	1.18	1.17	1.20
Run 3	1.17	1.21	1.17	1.17
Run 4	1.16	1.20	1.14	1.17
Run 5	1.16	1.21	1.16	1.18
Run 6	1.17	1.20	1.16	1.18
Run 7	1.16	1.20	1.18	1.19
Run 8	1.18	1.20	1.17	1.20
Run 9	1.18	1.19	1.17	1.17
Run 10	1.15	1.20	1.15	1.19
Average	**1.16** <	1.20	**1.16** <	1.18

Results (II)

Interval regression metrics

- Conformal methods achieve 95% coverage.
- CQR presenta menor amplitud media del intervalo que ICP.



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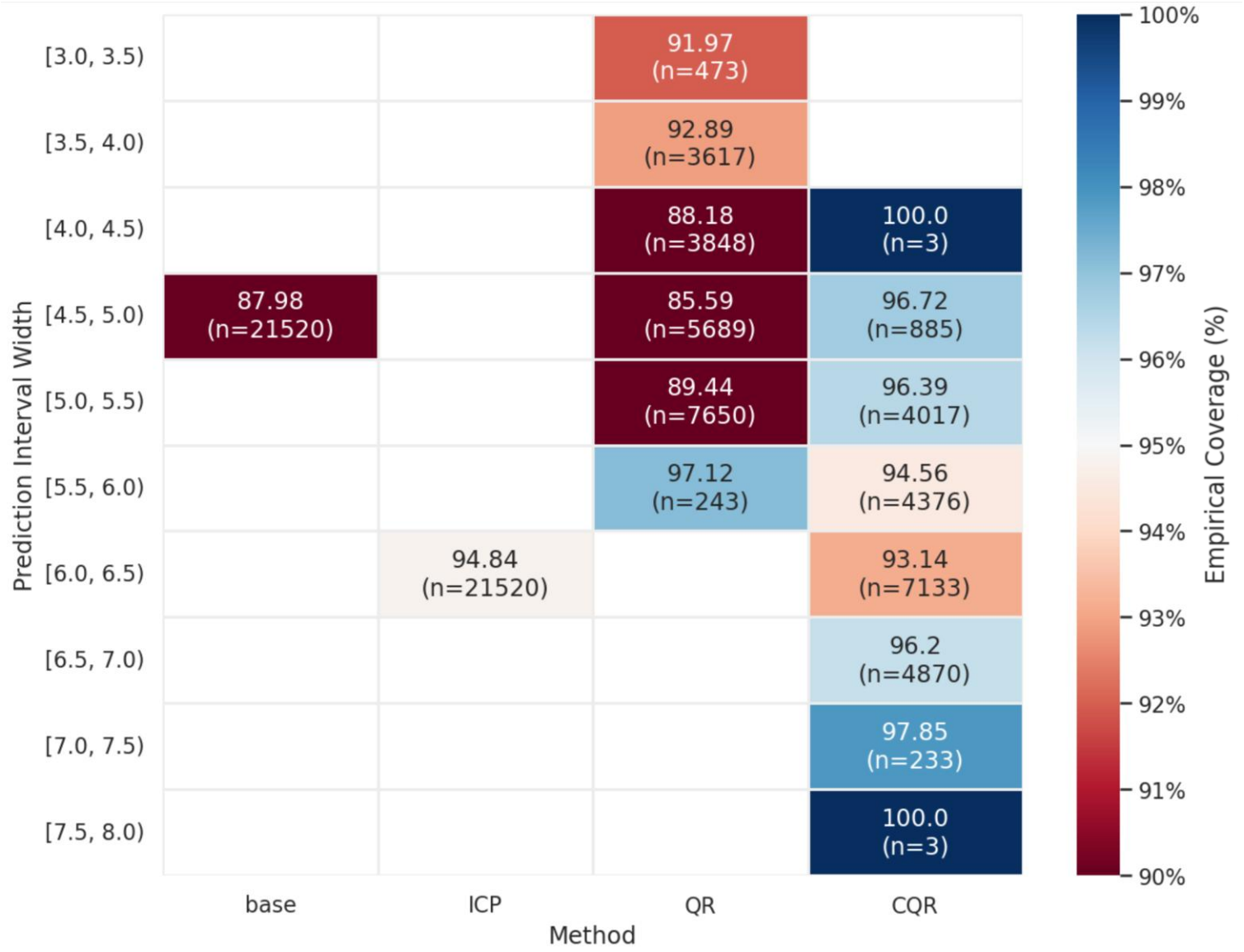
Results (III)

Coverage based on the width of the interval

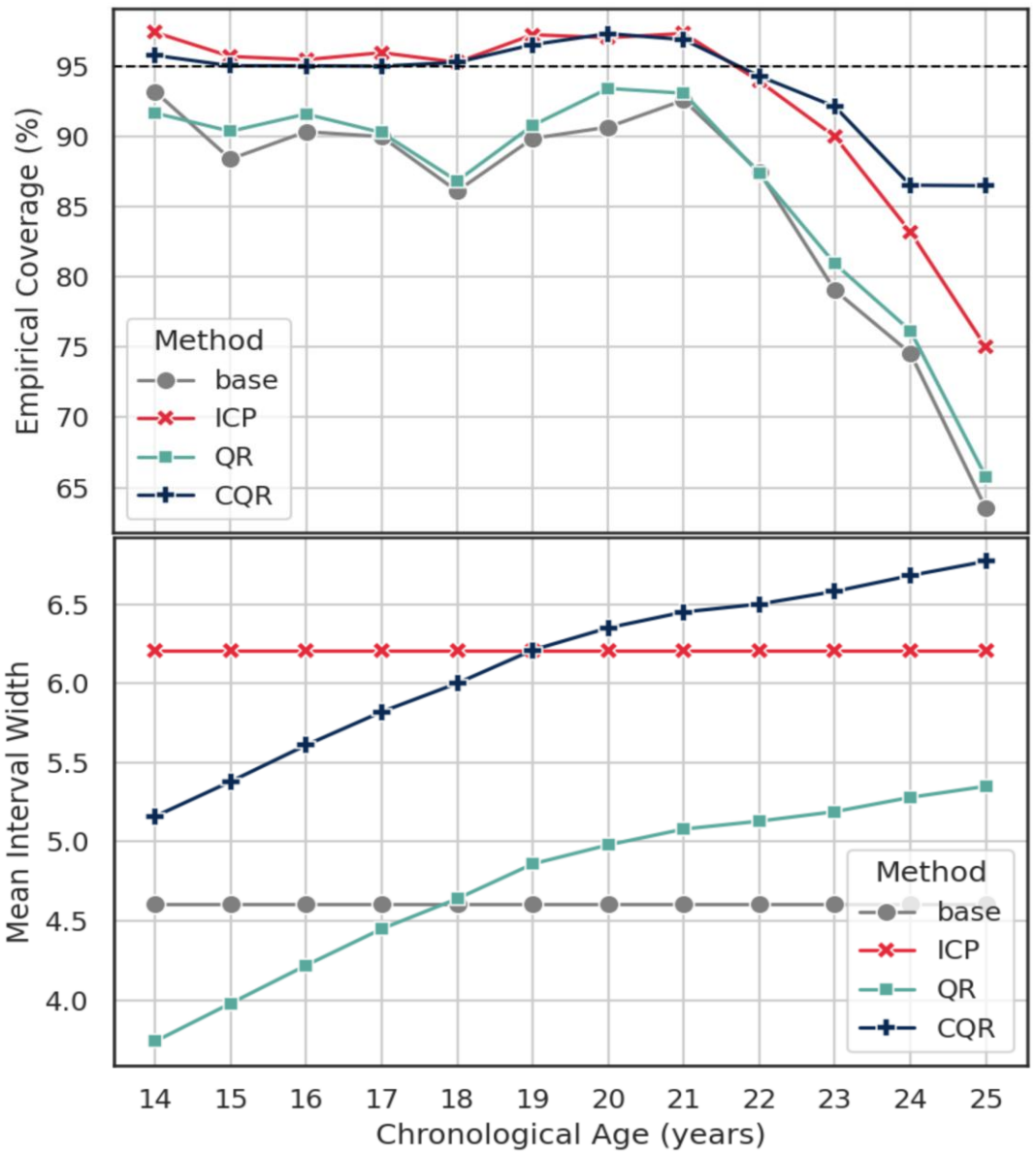
- Methods ‘base’ and ‘ICP’ always have intervals of the same width.
- CQR achieves good coverage in both narrowest and widest intervals, but its performance deteriorates in the intermediate ones.



Results (IV)

Coverage based on chronological age

- Coverage declines around the age of 22-23.
- CQR **provides the most stable coverage,** with its intervals becoming wider as the age increases.
- ICP reveals its weakness: all its intervals are the same size and fail to adapt to local uncertainty.



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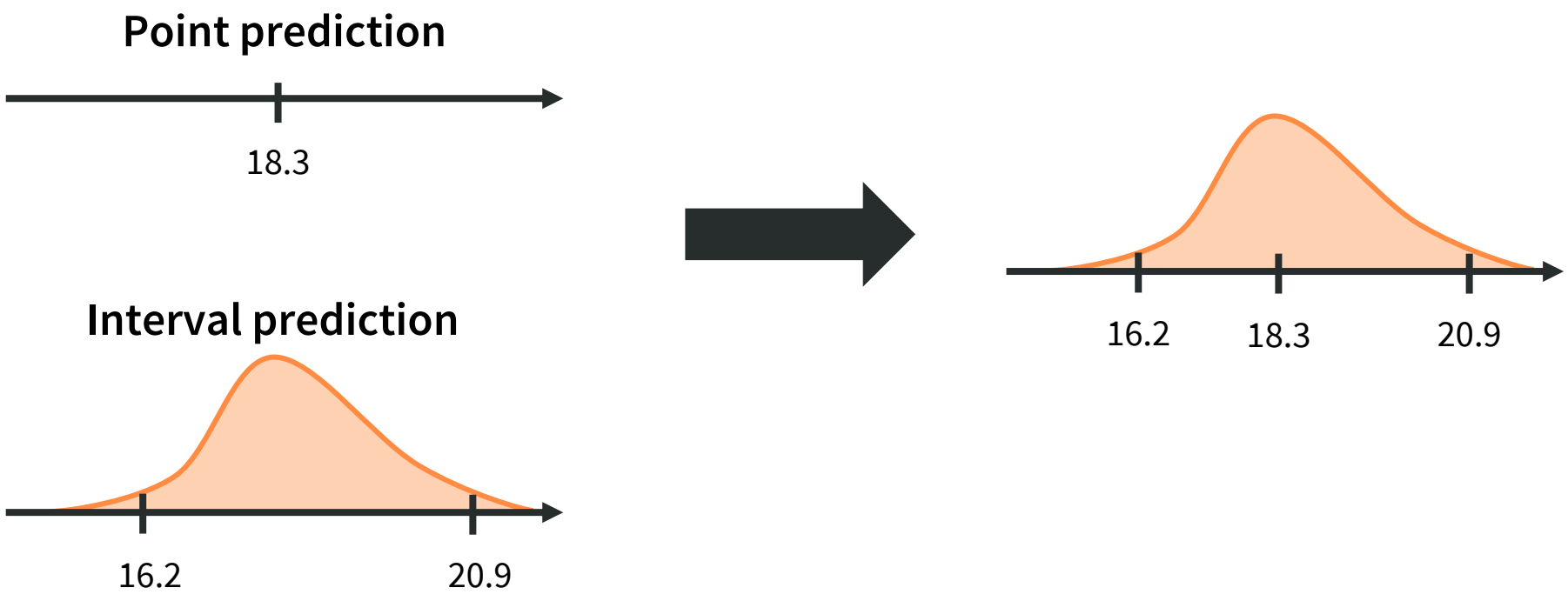
Conclusions (I)

1 Point and interval prediction are not mutually exclusive; they **complement each other**.



Conclusions (II)

2 Conformal prediction methods **do not always achieve the desired level of coverage across all subpopulations**, but there are differences between methods.

3 Adaptive methods **allow the size of the prediction interval to be adjusted dynamically**.

4 Although they require calibration data, conformal prediction **barely worsens the results of point prediction** in this problem.

5 Just as point prediction requires error analysis, interval prediction **requires a systematic assessment of coverage**.

Future work

1 Integration with other uncertainty quantification methods

For example, Conformalised Monte Carlo Dropout [1].

2 Leveraging domain expert knowledge

To carry out coverage analyses, taking into account advanced anthropological variables.

3 Combination with Out-of-Distribution (OOD) detection and explainability techniques

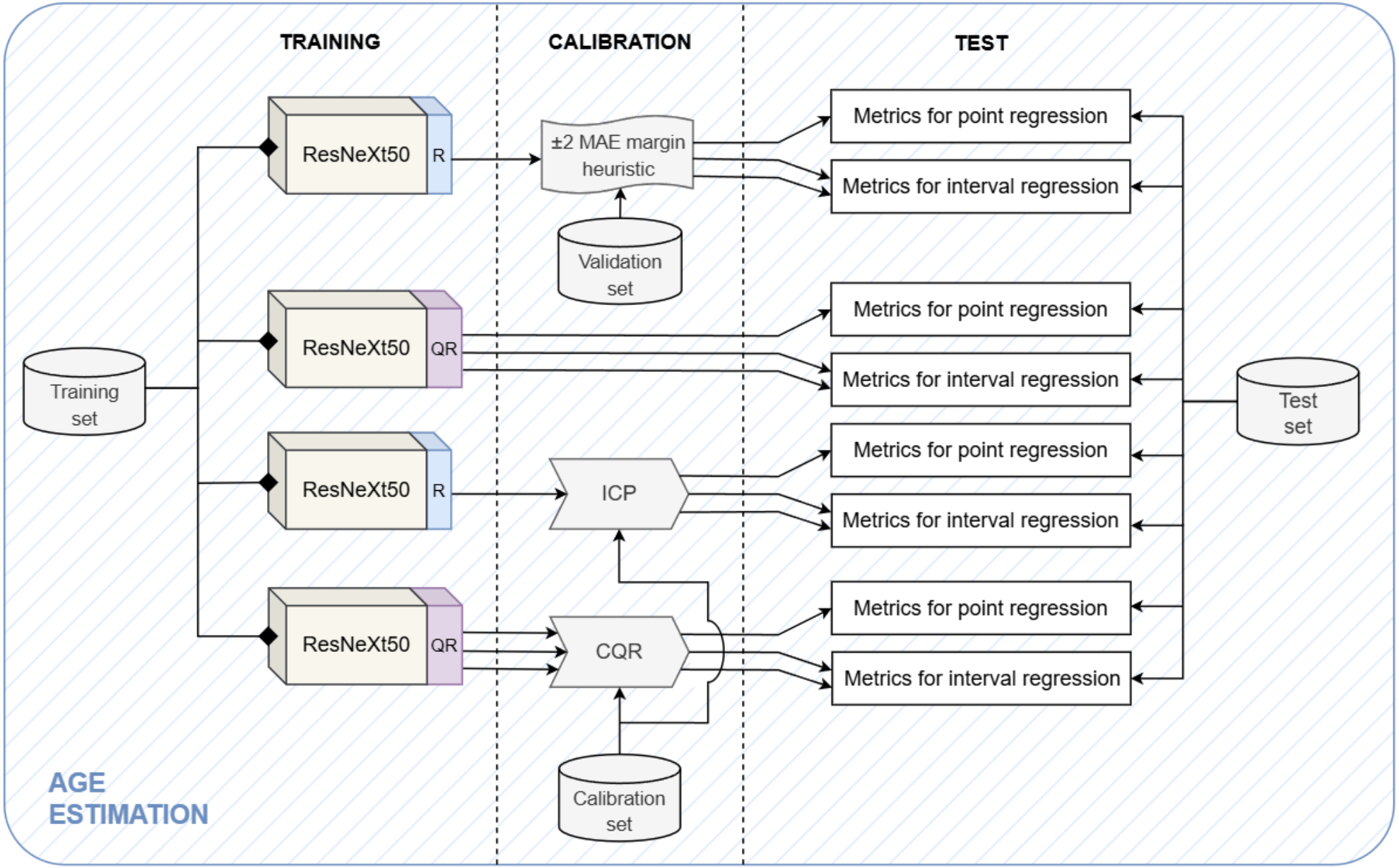
Addressing the shortcomings of conformal prediction.

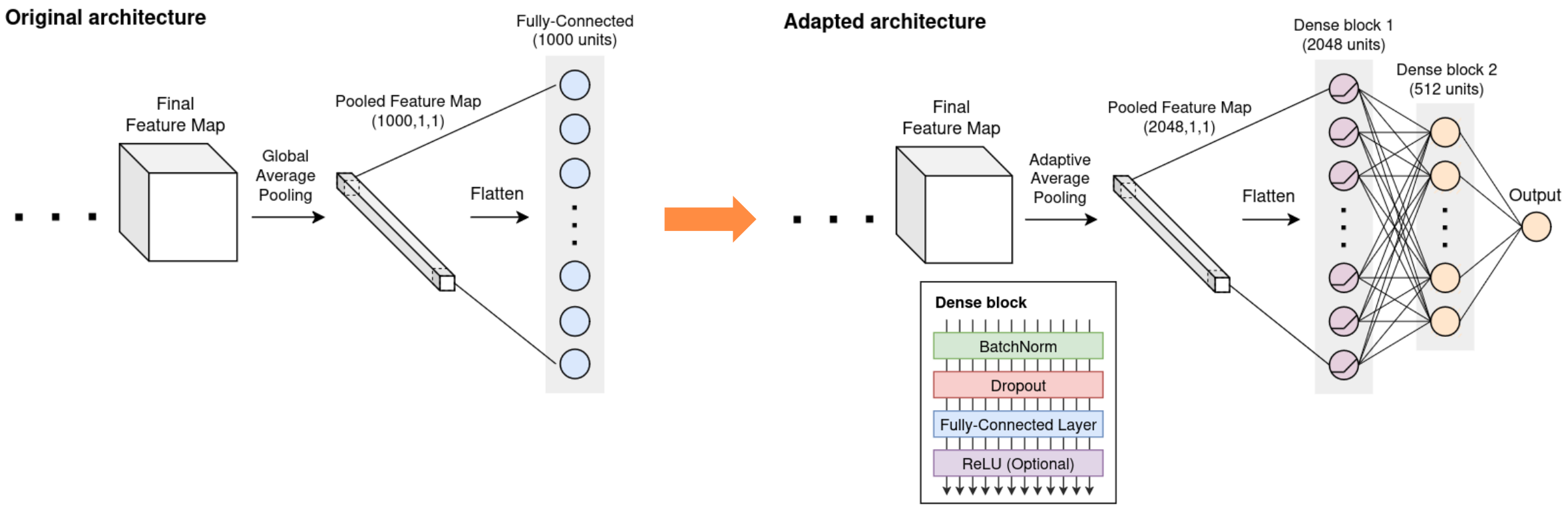
[1] Bethell, D., Gerasimou, S., & Calinescu, R. (2024, March). Robust uncertainty quantification using conformalised Monte Carlo prediction. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 38, No. 19, pp. 20939-20948).

Thank you for your attention.

Any questions?

Extras


TRAINING
CALIBRATION
TEST
ResNeXt50
R
±2 MAE margin heuristic
Validation set
Metrics for point regression
Metrics for interval regression
ResNeXt50
QR
Metrics for point regression
Metrics for interval regression
Training set
Test set
ResNeXt50
R
ICP
Metrics for point regression
Metrics for interval regression
ResNeXt50
QR
CQR
Metrics for point regression
Metrics for interval regression
Calibration set
AGE ESTIMATION

Selected architecture

ResNeXt50